

Linear Algebra

Friday 2 October 2026 | Main worksheet | Student copy - no solutions

Basics quick reference: 3-5 minutes + main route 50 minutes = 53-55 minutes. Supplements only if time remains.

1 Read an augmented matrix

Coefficients are left of the bar; the right-hand sides are right of it.

Keep the same variable order in every row. Each row represents one equation.

$$\begin{aligned}x + 2y &= 3 \\ y &= 4\end{aligned}$$

corresponds to

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 4 \end{array} \right]$$

2 Three legal elementary row operations

Apply an operation to the whole row, including the right-hand side.

Swap: $R_i \leftrightarrow R_j$. Scale: $R_i \leftarrow cR_i$, with $c \neq 0$.

Add a multiple of a different row: $R_i \leftarrow R_i + cR_j$, with $i \neq j$.

Each operation is reversible, so it preserves the solution set.

3 Leading entries and pivot positions

Read each nonzero row from left to right.

Its leading entry is its first nonzero entry. In echelon form, that position is a pivot position; the column containing it is a pivot column. Not every nonzero entry is a pivot.

4 REF and RREF: use this course convention

REF = row echelon form. RREF = reduced row echelon form.

REF: all zero rows are at the bottom; every leading entry is 1; pivots move strictly right as you go down; entries below each pivot are zero.

RREF: all REF conditions, and entries above each pivot are also zero.

Some textbooks allow other nonzero leading entries in REF; this course uses 1.

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2}$$

$$\left[\begin{array}{ccc} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array} \right]$$

REF (left) to RREF (right). These are ordinary matrices, not augmented matrices.

5 Count solutions: consistency first

After row reduction, read the equations before counting pivots.

In echelon form, free variables correspond to non-pivot variable columns.

A row $[0 \dots 0 \mid c]$ with $c \neq 0$ means $0 = c$: no solution.

If consistent and every variable column has a pivot: one unique solution.

If consistent and a variable column has no pivot: a free variable, hence infinitely many solutions over $\mathbb{R}$.

A zero row alone does not determine the answer. Not every column must have a pivot.

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Main route: L1-L6. Course convention: nonzero rows in REF have leading entry 1. Work over the real numbers.

L1 REF, RREF, or neither?

Week 2 tutor activity, p.5. Try alone: 90 seconds; then compare reasons.

For each matrix choose: RREF / REF but not RREF / neither. Circle the leading entries and explain your choice.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

A: _____ because _____

B: _____ because _____

C: _____ because _____

L2 Equations and free variables

Tutor-written activity. Try for 3 minutes. Not the hand-in question.

Write the system as $Ax = b$ and as an augmented matrix. Reduce it and describe every solution.

$$\begin{aligned} x + y + z &= 3 \\ 2x + 2y + 2z &= 6 \end{aligned}$$

Sizes: A is ____ $\times$ ____; unknown vector x has ____ entries; b has ____ entries.

Augmented matrix and row operations:

All solutions (include the range of any parameters):

Check: what changes if the right-hand side of the second equation is 7 instead of 6?

L3 has two working pages. Use 8 minutes for your first attempt, then compare and complete it during discussion.

Problem Sheet 2, Q2.2 (additional question). Not Q2.1, the hand-in problem.

$$\begin{array}{rcl} x_1 + x_2 + 2x_3 - 4x_4 & = & -1 \\ -x_1 + x_2 - 2x_3 - 2x_4 & = & -2 \\ -x_1 + 2x_2 - 4x_3 + x_4 & = & -1 \\ 3x_1 - 3x_4 & = & -3 \end{array}$$

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Continue L3. A correct route need not match the tutor's route. Finish by checking the original equations.

L3 Complete, justify, verify

Continuation of Problem Sheet 2, Q2.2. No extra timed task.

Continue your row reduction or write the resulting triangular system:

Back-substitution and solution (x_1, x_2, x_3, x_4) :

Explain why the solution is unique:

Substitute your result into all four original equations:

Reflection: which legal operation did you choose first, and why?

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Check consistency before counting free variables. Optional activities S1-S2 are in the separate supplement.

L4 How many solutions?

Tutor-written activity, related to PS2 Q2.13. Try for 3 minutes.

These are augmented matrices for real variables x, y . State the number of solutions; when consistent, write the whole solution set.

$$\mathbf{U} = \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array} \right]$$

$$\mathbf{N} = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array} \right]$$

$$\mathbf{F} = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array} \right]$$

U: number of solutions / solution set / reason:

N: number of solutions / solution set / reason:

F: number of solutions / solution set / reason:

L5 Challenge the claim

Tutor-written diagnostic. Think of a counterexample for 2 minutes.

“There is a zero row, so there are infinitely many solutions.” Is this always true? Give a counterexample. Explain why legal elementary row operations preserve the solution set.

L6 Exit check - without calculating

Tutor-written check. Write independently for 2 minutes.

1. What extra condition distinguishes RREF from REF?
2. What does a row $[0 \dots 0 \mid 1]$ mean?
3. What must you check before declaring infinitely many solutions?

